

#1

```
print("#1")
```

```
x,z=5,10
```

```
y=x+3
```

```
x=x-1
```

```
x=x+z
```

```
print('x:',x,'y:',y,'z:',z)
```

#2

```
print()
```

```
print("#2")
```

```
print(14//4,14%4,14/4)
```

#3

```
print()
```

```
print("#3")
```

```
print(2*'no'+3* '!')
```

```
print(2*('no'+3* '!'))
```

#4

```
print()
```

```
print("#4")
```

```
print(type(1+3))
```

```
print(type(1+3.0))
```

#5

```
print()
```

```
print("#5")
```

```
print(11+3)
```

```
print('11'+ '3')
```

```
#print('11'+3) will give error as it can concatct only str and  
not str with int
```

```
print()
```

```
print(type(5*2))
```

```
print(type(5**2))
```

```
print(type('5'+ '2'))
```

```
print(type('5'*2))
```

```
print(type(5/2))
```

```
print(type(5//2))
print(type(5%2))
print(type(5+2.0))
print(type(5.0*2.0))
#print(type('5'-2))will give error as it does not support(-) with
str
```

#5a

```
print()
print("#5a")
x,y,z,q=5,5,6,6
print(x>y)
print(y<=z)
#print(z<>x) it dosen't support <>
print(z==q)
print(x<y>z)
#print(x==y<>z) it dosen't support <>
```

#5b

```
print()
print("#5b")
a,b,c=3,4,5
print(a or b)      # or will return the 1st value (3)
print(b and c)     # and will return the 2nd value(5)
print(a and not b)  # not returns false
print(not c or not b)  # not returns false
```

#5c

```
print()
print("#5c")
ch,i,fl=5,2,4
db,fd=5.0,36.0
A=(ch+i)/db
B=fd/db*ch/2
print(A)
print(B)
```

#6

```
print()
print("#6")
a,b=3,6
c=b/a
```

```
print(c)
c=b//a
print(c)
c=b%a
print(c)
```

#7

```
print()
print("#7")
p,q,y=1,2,3
print(p>q<y)
a,N,b=1,2,3
print(a<=N<=b)
```

#7a

```
print()
print("#7a")
print(25/5 or 2.0+20/10)
```

#7b

```
print()
print("#7b")
print(a or (b and (not c)))
```

```
print((p and q)or (not r))
```

#7c

```
print()
print("#7c")
print((5<10)and (10<5) or (3<18) and not 8<18)
```

```
print((5<10) or (50<100/0))
```

#8

```
print()
print("#8")
x,y=6,3
age=21
state='Goa'
```

```
name='Subhrajit'
print(x%y==0)
print(age>=18 and state=='Goa')
print(name!='Nimra')
```

#9

```
print()
print("#9")
import math
a,b,c=3,4,5
p,q,r=7.0,9.3,10.51
x,y,z=25.519,10-24.113,231.05
print(math.pow(a/b,3.5))
print(math.sin(p/q)+math.cos(a-c))
print(x/y+math.floor(p*a/b))
print(((math.sqrt(b)*a)-c))
print(((math.ceil(p)+a)*c))
```

#10

```
print()
print("#10")
import math
a,b,c=2,3,4
x,y=30,2
p,q,r,s=3,4,1,2
print(math.sqrt(a*a+b*b+c*c))
print(2-y*math.exp(2*y)+4*y)
print(p+q/math.pow((r+s),4))
print((math.cos(x)/math.tan(x))+x)
print(math.fabs(math.exp(2)-x))
print(math.exp(4))
```

FILE : CH7EG1.PY

```
string='#'
pattern=""
for a in range(5):
    pattern+=string
    print(pattern)
```

FILE : CH7EG18.PY

```
a=10
b=23
c=23
print(a<b)
print(b<=c)
print(a<b<=c)
```

FILE : CH7EG22.PY

```
print(10>2)
print(2<4)
print((2+5,6+6))
print(5>2 and not (10>11))
#print([2,4[3,4]]) subscript with out array name    still it
would be wrong
print(4==2**2)
print(4,2**2)
print('cat'<'dog')
print('Cat'<'dog')
print(5>2 and 10>11)
print(5>2 or 10>11)
```

FILE : CH7EG25.PY

```
import math
r=7.5
area=4*math.pi*r*r
volume=4/3*math.pi*math.pow(r,3)
print("radius of the sphere:",r,"meter")
print("area of the sphere:",area,"unit square")
print("volume of the sphere:",volume,"unit cube")
```

FILE : CH7EG28.PY

```
import random
import math
fnum=random.random( )*(95-45)+45
inum=math.ceil(fnum)
print("random number between 45..95:")
print(fnum)
print("nearest higher integer:",inum)
```

FILE : CH7EG29.PY

```
import random
num1=random.randint(450,950)-450
num2=random.randint(450,950)-450
avg=(num1+num2)/2
print("random integer in the range 450 to 950", num1,num2)
print("their average :",avg)
```

FILE : CH7EG30.PY

```
import random
num1=random.randrange(10,70,14)
num2=random.randrange(10,70,14)
num3=random.randrange(10,70,14)
set1={num1,num2,num3}
print("random integers in the range 10...70,stepvalue 14
:",num1,num2,num3)
print("set created :", set1)
```

FILE : CH7EG32.PY

```
import math
a=int(input("Enter he length of side a :"))
b=int(input("Enter he length of side b :"))
c=int(input("Enter he length of side c :"))
s=(a+b+c)/2
area=math.sqrt(s*(s-a)*(s-b)*(s-c))
print("area of a triangle",area)
```

FILE : CH7EG31.PY

```
import statistics as stat
list1=[22,13,28,13,22,25,7,13,25]
list_mean= stat.mean(list1)
list_median= stat.median(list1)
list_mode= stat.mode(list1)
print("Given list :",list1)
print("mean: list :",list_mean)
print("median: list :",list_median)
print("mode: list :",list_mode)

print()
list1=[5,6,7,5,6,5,5,9,11,12,23,5]
list_mean= stat.mean(list1)
list_median= stat.median(list1)
list_mode= stat.mode(list1)
print("Given list :",list1)
print("mean: list :",list_mean)
print("median: list :",list_median)
print("mode: list :",list_mode)
```